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ChE 210 | Quiz #2 | 30 September 2019

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Pure methane (CH_4) enter a reactor at 16 kg/hr along with 300 kg/hr of air (21 mol% O_2 and 79 mol% N_2). A single product stream contains no methane and only CO_2, O_2, N_2, and H_2O. Assume steady state.

(a) Draw a labeled block diagram and calculate the degrees of freedom for the process using p. 105 of the text. Be sure to give values for 4a, 4b, 5a, 5b, 5c, and 5d as indicated on page 105.

(b) Calculate the molar flow rates of CH_4, O_2, and N_2 entering the reactor in kmol/hr.

(c) Write and solve the appropriate mole balances to find the molar flow rates of all the species exiting the reactor in kmol/hr.

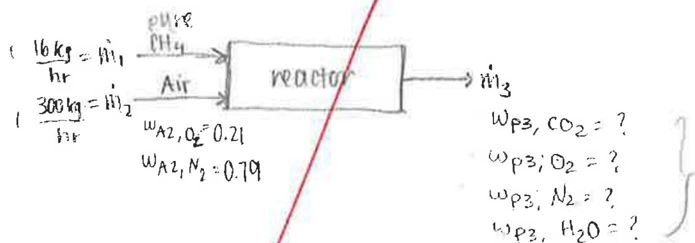
(d) What is the composition of the product in mole fractions?

$CH_4 = 12.01 + 4(1.008) = 16.042$ kg/kmol

$O_2 = 2(16.00) = 32$ kg/kmol

$N_2 = 2(14.01) = 28.02$ kg/kmol

a)



Assumption: steady state $\frac{dm}{dt} = 0$

4a. 7
4b. ~~2~~
9

5a. 2
5b. ~~2~~
5c. 0
5d. 5
9

DOF = 9 - 9 = 0

b) $\dot m_1 w_{m1} = 16\frac{kg}{hr} \cdot 1 = 16\frac{kg}{hr} CH_4 \Rightarrow \dot n_1 = 16\frac{kg}{hr} CH_4 \times \frac{1\ kmol}{16.042\ kg\ CH_4} = 0.997 \frac{kmol}{hr} CH_4$

$\dot m_2 w_{A2,O_2} = 300\frac{kg}{hr} \cdot 0.21 = \cancel{63}\frac{kg}{hr} O_2 \Rightarrow \dot n_{A2,O_2} = 63\frac{kg}{hr} O_2 \times \frac{1\ kmol}{32\ kg\ O_2} = 1.969 \frac{kmol}{hr} O_2$

$m_2 w_{A2,N_2} = 300\frac{kg}{hr} \cdot 0.79 = \cancel{237}\frac{kg}{hr} N_2 \Rightarrow \dot n_{A2,N_2} = 237\frac{kg}{hr} N_2 \times \frac{1\ kmol}{28.02\ kg\ N_2} = 8.458 \frac{kmol}{hr} N_2$

c) General Mass Balance [Mass is conserved]

$\dot m_1 + \dot m_2 = \dot m_3$

$16\ kg + 300\ kg = \dot m_3$

$\dot m_3 = 316\frac{kg}{hr}$ OK

Material Balance

O_2

$\dot m_2 w_{A2,O_2} = \dot m_3 w_{P3,O_2}$

$63\frac{kg}{hr} = 316\frac{kg}{hr} \cdot w_{P3,O_2}$

$w_{P3,O_2} = 0.199\ kg\ O_2$

$n_{wP3,O_2} = 0.199\ kg\ O_2 \times \frac{1\ kmol}{32\ kg\ O_2} = 0.00623$

N_2: $\dot m_2 w_{A2,N_2} = \dot m_3 w_{P3,N_2}$

$237\frac{kg}{hr} = 316\frac{kg}{hr} \cdot w_{P3,N_2}$

$w_{P3,N_2} = 0.75\ kg\ N_2$

$n_{wP3,N_2} = 0.75\ kg\ N_2 \times \frac{1\ kmol}{28.02\ kg\ N_2} = 0.0268$

CO_2

$0 = m_3 w_{P3,CO_2}$

$\dot m_3(1 - 0.199 - 0.75 - w_{CO_2} - w_{H_2O})$

H_2O

$0 = m_3 w_{P3,H_2O}$